



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

MYOCARDIAL INFARCTION/ IHD

CARDIOVASCULAR SYSTEM/PPOM 2

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Do.
Make.
Heal.
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Session Objectives

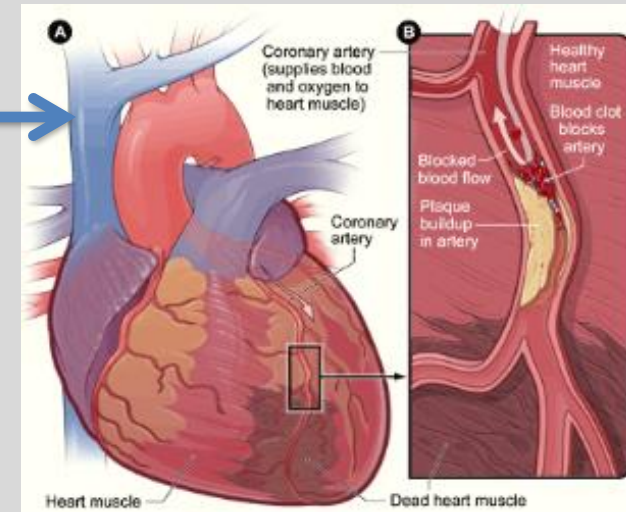
- 1. Describe and discuss the general concepts of **ischemic heart disease (including the epidemiology and risk factors)** and **define and compare stable, unstable, and Prinzmetal angina (review)**.
- 2. Explain the **pathogenesis/mechanism of coronary thrombosis and how it leads to myocardial infarction** and **delineate the initial changes of myocardial necrosis at the cellular and subcellular levels**.
- 3. Discuss, describe, compare, and differentiate the **changes of myocardial infarct as the time sequence evolves in morphological and pathophysiological terms**.
- 4. Correlate **myocardial infarction changes with clinical and laboratory manifestations**.
- 5. Discuss, describe, and evaluate **complications of myocardial infarctions including arrhythmias, pericarditis, ventricular free wall rupture, true and false aneurysms, papillary muscle and septal rupture, and contractile dysfunction**.

Outline

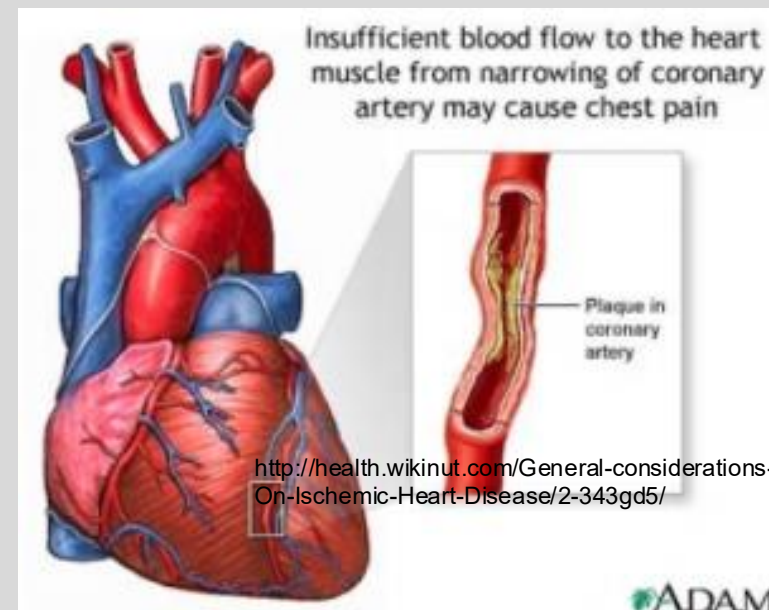
- Review IHD
- Overview of Myocardial Infarction
 - Clinical: Symptoms
 - EKG/ECG changes
 - Biochemical markers
- Pathophysiology of Cellular Injury
- Gross and microscopic changes in MI
- Complications
- Practice questions

IHD - Syndromes

- Myocardial infarction, most significant form, where ischemia causes death of cardiac muscle
- Angina pectoris, where ischemia isn't severe enough to cause infarction, but is often precursor
- Chronic IHD with heart failure
- Sudden cardiac death



<http://www.nhlbi.nih.gov/health/health-topics/topics/cad/signs>



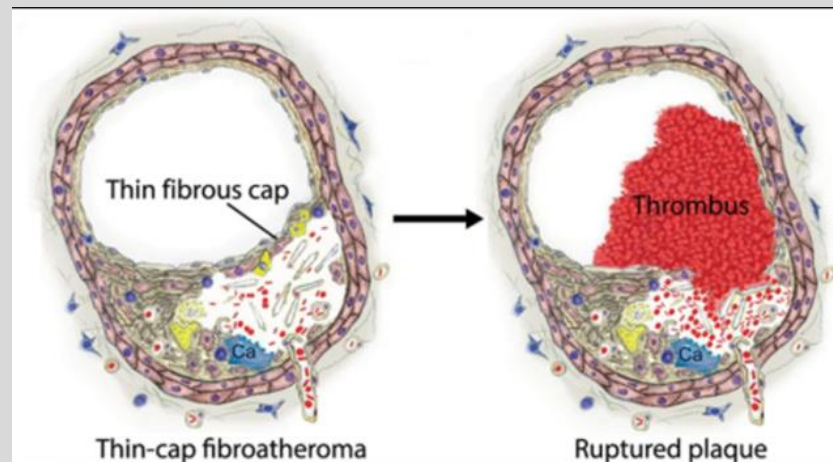
<http://health.wikinut.com/General-considerations-On-Ischemic-Heart-Disease/2-343gd5/>

IHD – Consequences of Myocardial Ischemia

- Stable angina – Previously discussed, chest pain with physical activity
- Unstable angina – Previously discussed, chest pain at rest
- Prinzmetal angina – vasospasm- does not depend on exercise or rest
- Myocardial infarction – acute plaque change with total thrombotic occlusion, followed by myocardial death
- Sudden cardiac death – often due to a disrupted plaque producing ischemia which induces a fatal ventricular arrhythmia

IHD - Pathogenesis

- Over 90% of patients with IHD have atherosclerosis within one or more coronary arteries
- Obstructions occupying 75% or more of the lumen cause symptoms induced by exertion
- 90% obstruction can lead to insufficient coronary blood flow even at rest
- Acute coronary syndrome
 - #1 source of MI is acute plaque rupture of vulnerable or unstable plaque with thrombus formation
 - Occurs via rupture, erosion, ulceration, and/or deep hemorrhage
 - Often associated with intralesional inflammation, which mediates plaque disruption



Myocardial Infarction

- Necrosis of cardiac myocytes (Irreversible)
- Usually due to rupture of an atherosclerotic plaque with thrombus - complete occlusion of coronary artery
- Leading cause of death in U.S

Myocardial Infarction

- Initial phase:
Subendocardial necrosis involving
<50% of myocardial
thickness
 - EKG: **ST-segment depression**
- Continued or severe
ischemia leads to
Transmural necrosis
involving most of
myocardial wall
 - EKG: **ST-segment elevation**



Myocardial Infarction

- Transmural MI =
 - ischemic necrosis involves full thickness of ventricle wall (in distribution of single coronary artery)
 - most MIs are transmural
 - associated with coronary atherosclerosis, acute plaque change, superimposed thrombosis
- Subendocardial infarct
 - area of ischemic necrosis limited to inner one third or less than half of ventricle wall
 - (n.b. the subendocardial zone is the least well-perfused area of the myocardium - most vulnerable to reduction in coronary blood flow)

Clinical features

- Severe, crushing chest pain lasting greater than 20 minutes
- Radiation to neck jaws, arms, back commonly left side
- Other symptoms include diaphoresis, dyspnea, weakness or fatigue, nausea and vomiting, sense of impending doom, syncope
- Symptoms not relieved by nitroglycerin
- Silent MI's may occur in individuals with diabetes and elderly due to pain threshold or problems with the nervous system

Frequency and Sites of Myocardial Infarction

- Occlusion **Left Anterior Descending Artery (LAD)**: infarction of LV anterior wall, anterior septum and apex circumferentially
 - *LAD is most commonly involved artery in MI (40-50%)*
- Occlusion **Right Coronary Artery (RCA)**: infarction of posterior wall, posterior septum, and papillary muscles of LV
 - *RCA is 2nd most commonly involved artery in MI (30-40%)*
- Occlusion: **Left Circumflex Artery**: infarction of lateral wall of LV except apex (15-20%)



1–14 hrs
after MI
onset



12–24 hrs
after MI
onset



Days to
months



Phase 1:
ST elevation
Hyperacute wide/large T

Phase 2:
ST elevation persists
Q starts to develop
T starts to invert

Phase 3:
ST elevation subsides
Q becomes deeper
T inversion becomes deeper
QT prolongs

Acute STEMI
→ Emergent reperfusion if
persistent chest pain or last
chest pain within 12 (24) hrs

Recent or subacute STEMI
→ Reperfusion with PCI may be
indicated for persistent/recurrent
chest pain, but is not emergent
from the ECG standpoint. Fibrinolytics
are not indicated.

This pattern may also be seen after less prolonged,
transient ischemia without extensive infarction
(e.g., 1–2 hrs), where PCI is beneficial

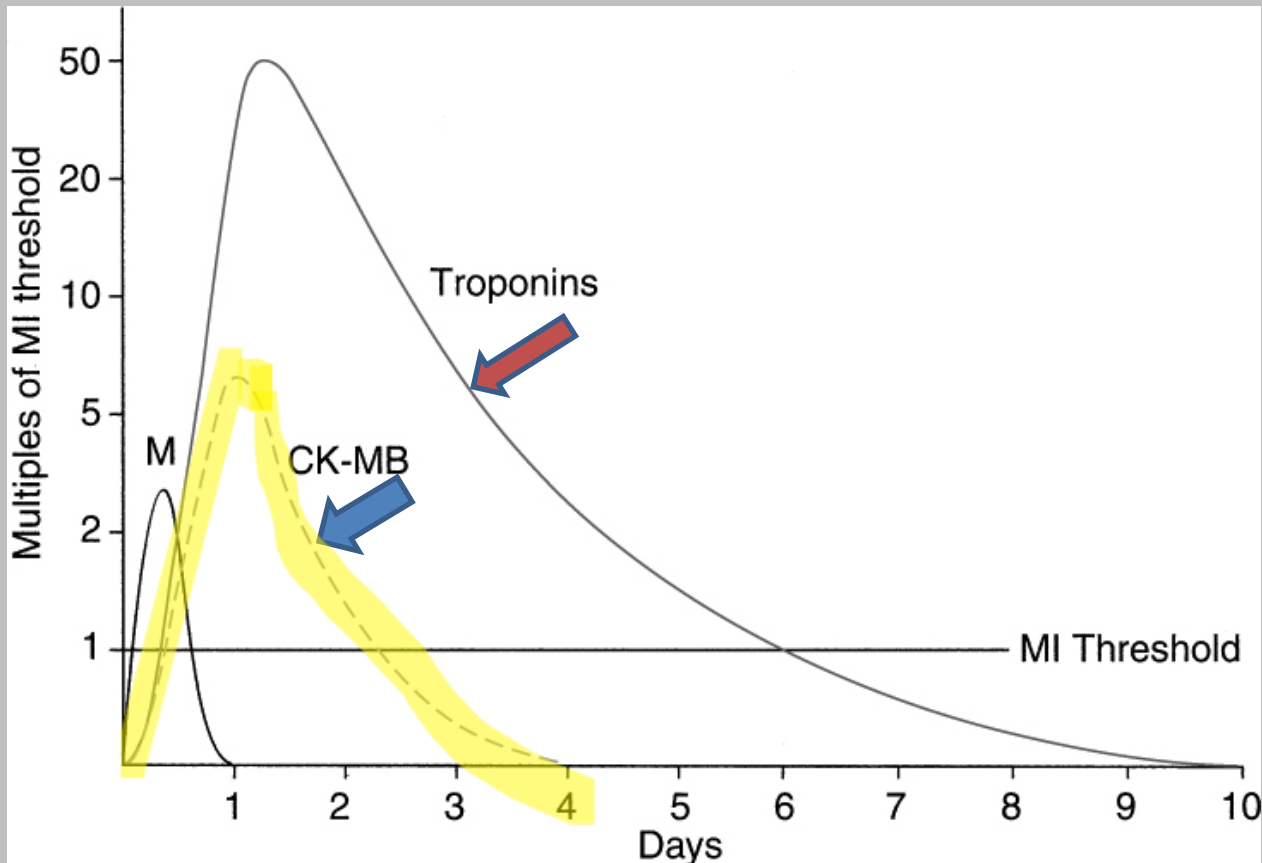
Old MI

<https://thoracickey.com/st-segment-elevation-myocardial-infarction-6/>

MI Diagnosis

- Clinical: Symptoms
- EKG/ECG changes
- Biochemical markers: myoglobin, troponins T and I, creatine kinase (CK-MB), lactate dehydrogenase
- Troponins: proteins that regulate Ca-mediated contraction of cardiac and skeletal muscle.
 - **Tn-I and Tn-T are cardiac specific** and rise 2 to 4 hours after an acute MI (up to 12 hours)
 - peak at 24 - 48 hours
 - **remain elevated for 7 to 10 days** Troponins elevated longer
- **CK-MB** also rises within 4 hours (up to 8) of an acute MI
 - peaks at 24 hours
 - **returns to normal by 72 hours**
- CK-MB is useful for detecting re-infarction that occurs days after an initial MI

Laboratory Enzymes



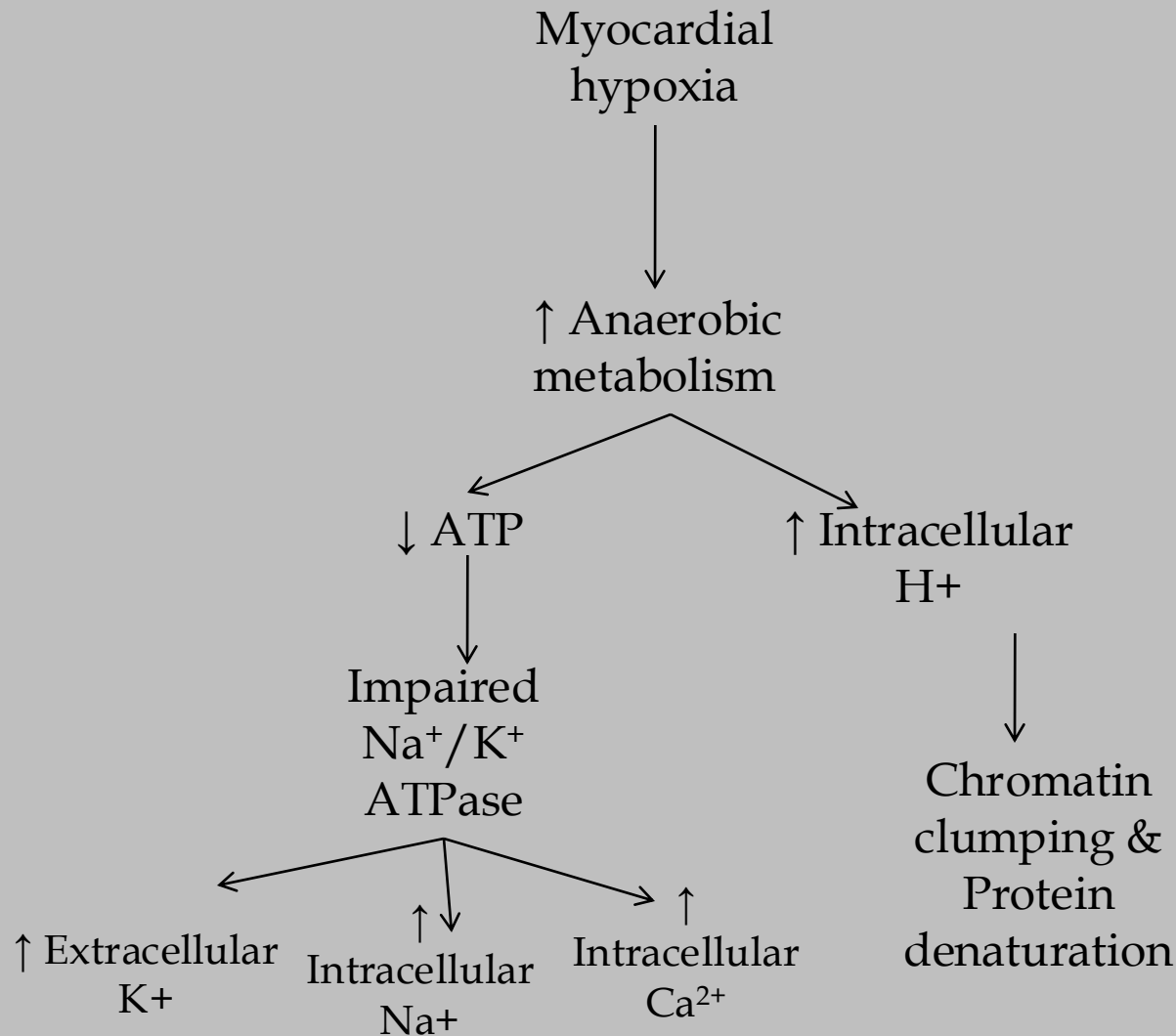
Review of Cellular Injury

- Cellular injury: stress exceeds cell's ability to adapt
- Hypoxia or low oxygen delivery to tissue = cellular injury
- Causes of Hypoxia include
 - Ischemia – decrease blood flow through an organ
 - Hypoxemia – low partial pressure of oxygen in blood
 - Decreased oxygen carrying capacity – Hemoglobin loss or anemia

MI, continued

- Early biochemical consequence myocardial ischemia
end of aerobic glycolysis within seconds
 - accumulation of noxious breakdown productions (e.g. lactic acid because of increase in anaerobic glycolysis)
 - loss of contractility occurs within 60 seconds of ischemia
- Early changes are still reversible (EM changes)
 - it takes at least 20 to 40 minutes for irreversible damage (necrosis) to cardiac myocytes.
 - What does this imply clinically?

Impact of Oxygen Deprivation



Reversible and Irreversible Cellular Injury

- **Initial phase of injury: reversible**
 - Hallmark is **cellular swelling**
 - Cytosol swelling results in loss of microvilli and membrane blebbing
 - Swelling of RER results in dissociation of ribosomes and decreased protein synthesis
- Further damage becomes **irreversible** and hallmark is **membrane damage**
 - Plasma membrane, mitochondrial membrane, lysosome membrane
 - End result of irreversible injury is cell death

Myocardial Infarction Changes

TIME	GROSS	MICRO
0 – 30 minutes; Reversible	None	None
30 min – 4 hours	None	Early “ wavy fibers ”
4 – 12 hours	Focal mottling	Start of coagulative necrosis , edema, hemorrhage
12 – 24 hours	Dark mottling	Coagulative necrosis , pyknosis of nuclei, myocyte hypereosinophilia, contraction band necrosis, start of neutrophil infiltrate
1 – 3 days	Mottling with yellow-tan infarct center	coagulative necrosis with loss of nuclei and striations, interstitial infiltration of neutrophils

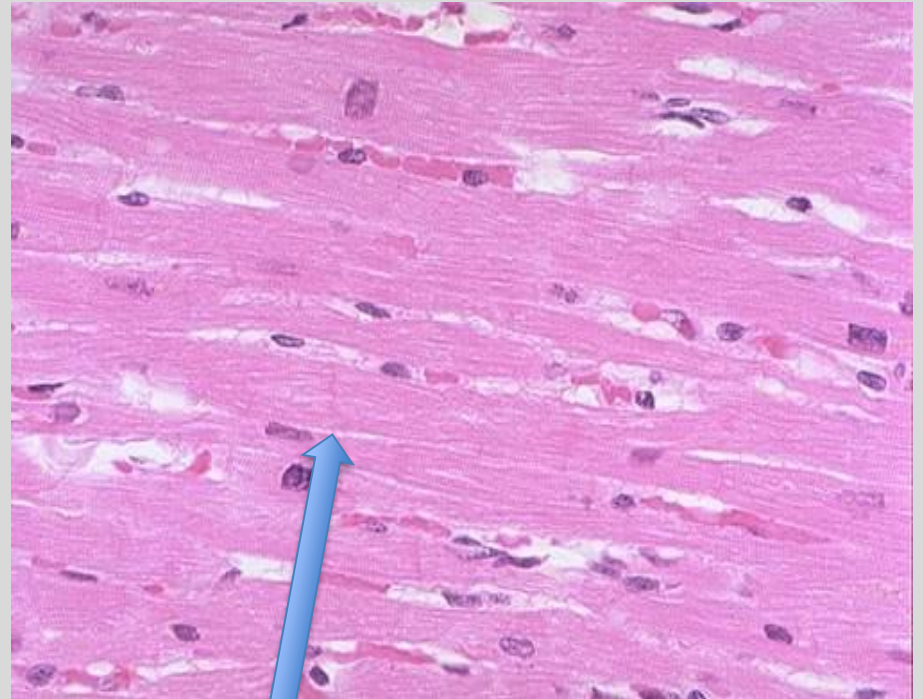
Myocardial Infarction Changes

TIME	GROSS	MICRO
3 – 7 days	Hyperemic border, central yellow-tan softening	Start of disintegration of dead myofibers, early phagocytosis of dead cells by macrophages
7 – 10 days	Maximal yellow-tan, soft with depressed red-tan margins	Phagocytosis well under way; early granulation tissue
10 – 14 days	Red-grey depressed infarct borders	Granulation tissue well formed with angiogenesis, collagen
2 – 8 weeks	Grey-white scar, from border to center of infarct	Increased collagen , decreased cellularity/vascularity
> 2 months	Scarring complete	Dense collagen scar

Robbins, Pathologic Basis of Disease, 10th edition, 2020, pg. 547, Table 12-5

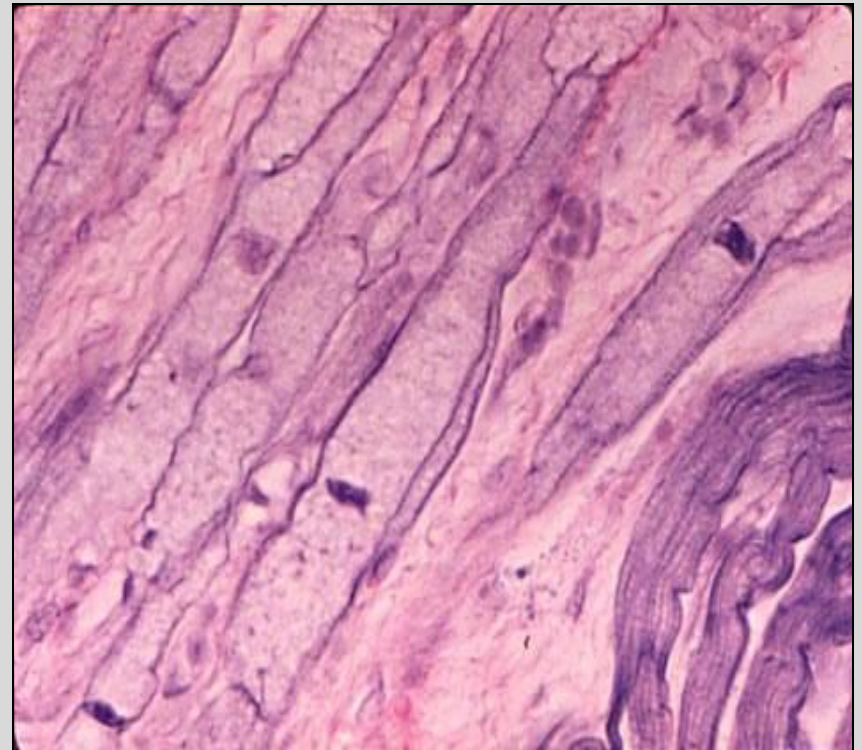
Normal Myocardium

- Central nuclei and syncytial arrangement of myocardial fibers
- Pale pink intercalated disks - help synchronize myocyte contraction via electrical coupling of adjoining cells.

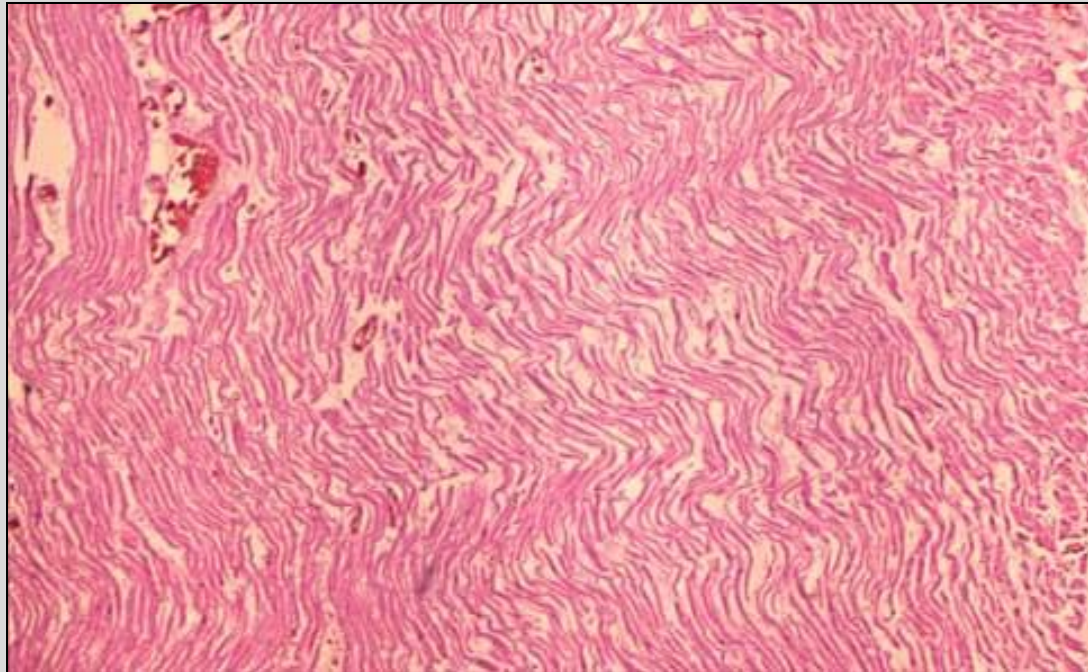


Cellular Swelling: 0 to 30 minutes

- Acute and Reversible
- Ischemia causes decreased oxidative phosphorylation and decreased ATP
- Decreased function of Na^+ pump and subsequent influx of H_2O and Na^+ resulting in cell swelling



Wavy fibers



- 30 minutes to about 4 hours

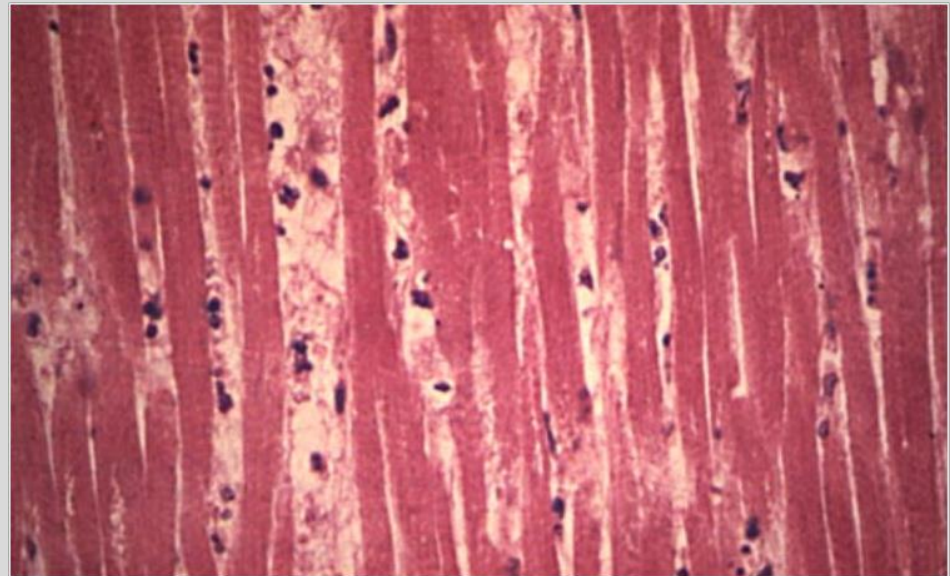
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<http://library.med.utah.edu/WebPath/webpath.html>

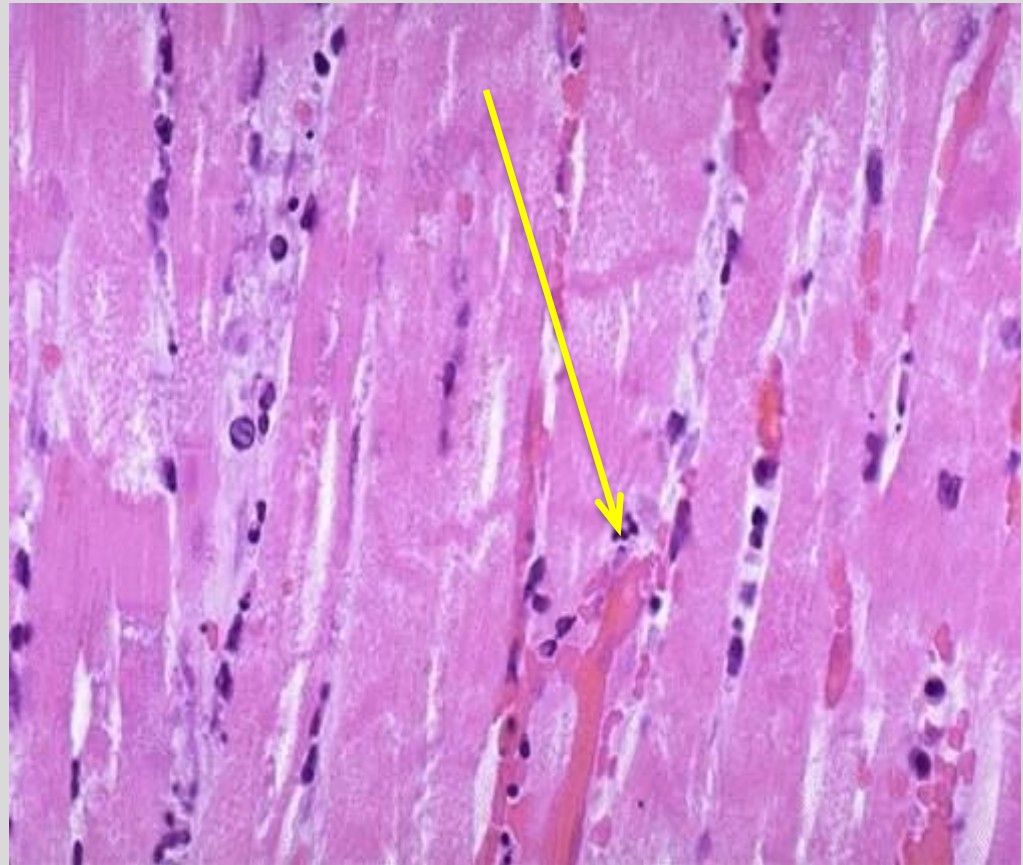
Early Coagulative Necrosis

- Edema – 4 to 12 hours
- Usually cannot see gross changes in heart if MI is less than 12 hours old
- Unless aided by use of TTC which stains normal myocardium brick-red and leaves infarcted area pale and unstained (loss of dehydrogenases)



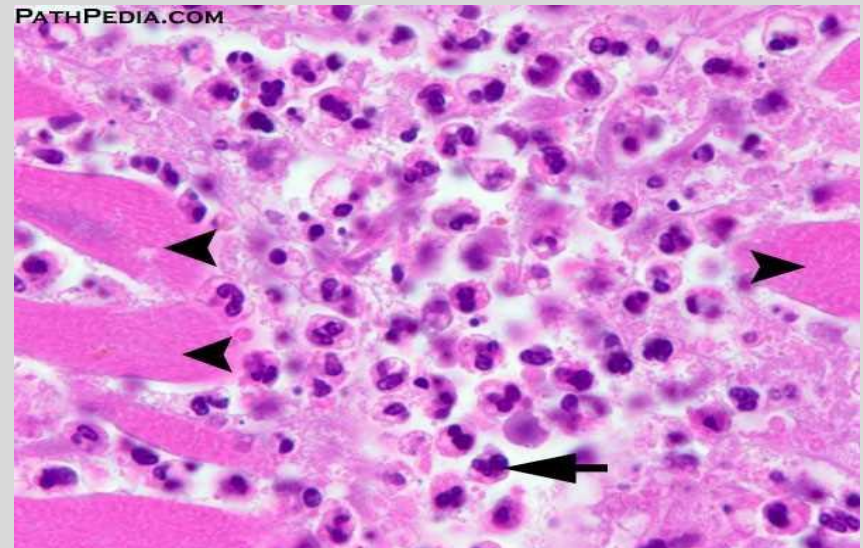
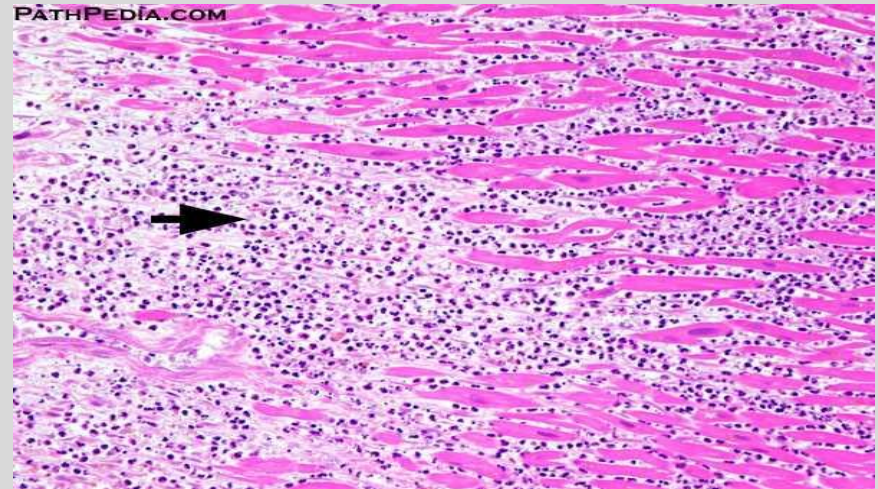
Early Acute MI: 24 hours

- Increased loss of cross striations, some contraction bands are also seen, and the nuclei are undergoing karyolysis.
- Some neutrophils beginning to infiltrate the myocardium.



Acute MI: 1-3 days

- Coagulative necrosis with loss of nuclei and striations,
- Interstitial infiltration of **neutrophils**



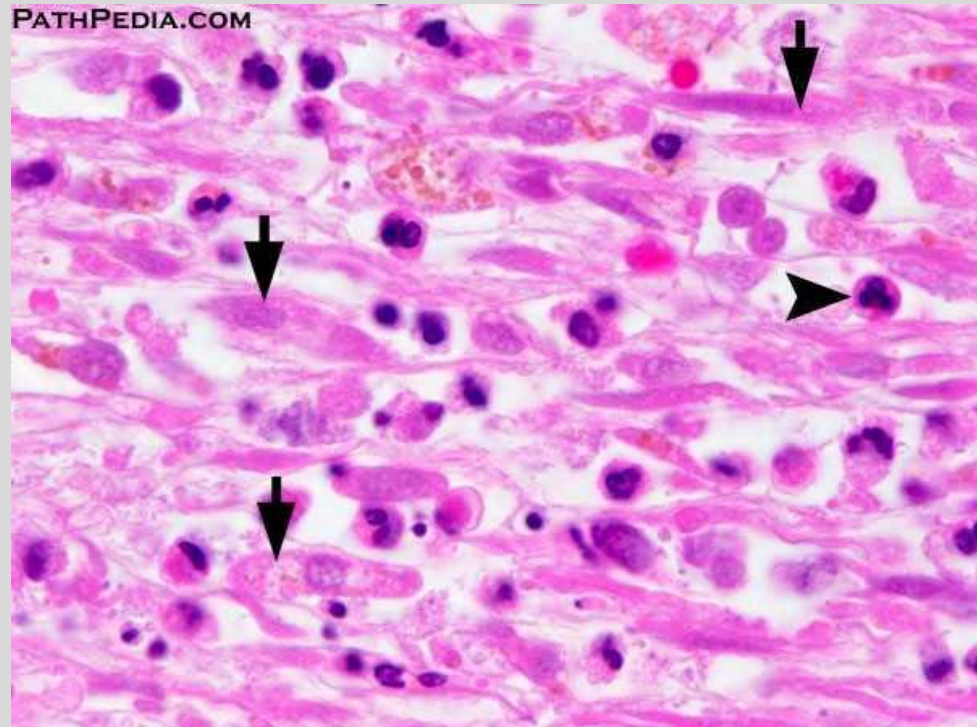
Acute MI: 1-3 days

- Acute myocardial infarction in the septum.
- After several days, **yellowish center** with necrosis and inflammation surrounded by hyperemic border.



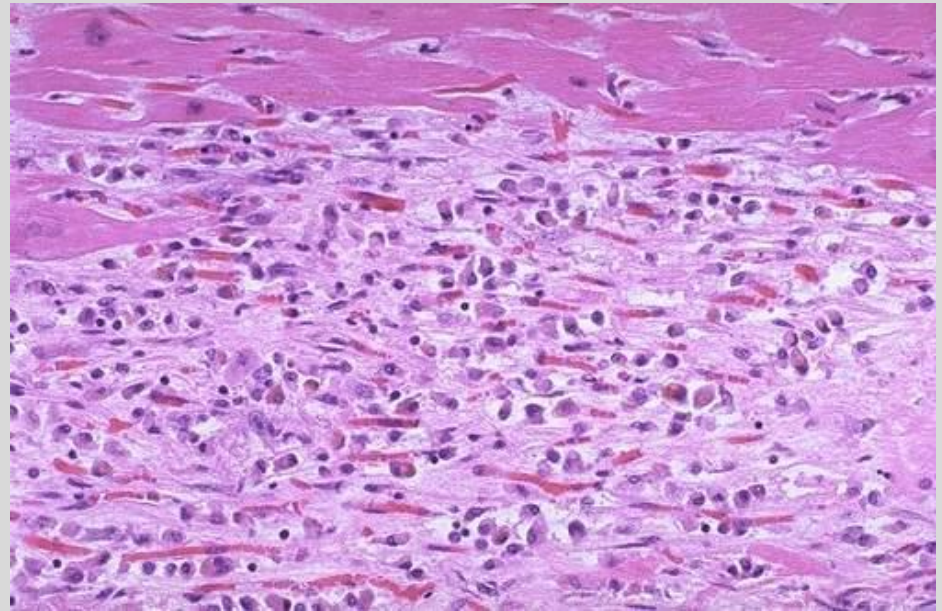
Acute MI: 3-7 days

- Start of disintegration of dead myofibers,
- early phagocytosis of dead cells by **macrophages**



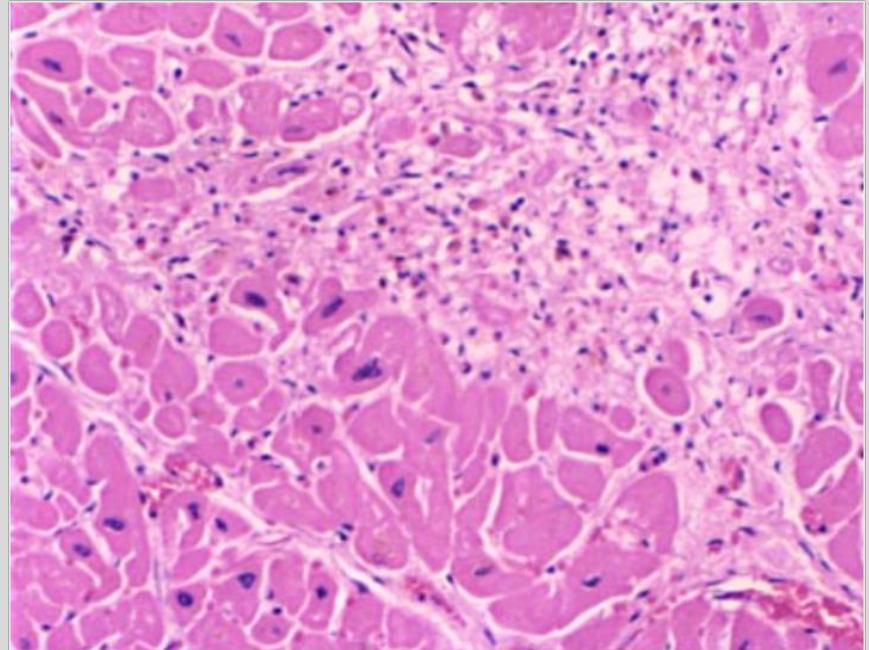
Acute MI: 7-10 days

- Below myocardial fibers (top) are many macrophages along with numerous capillaries and little collagenization
- Early **granulation tissue** formation

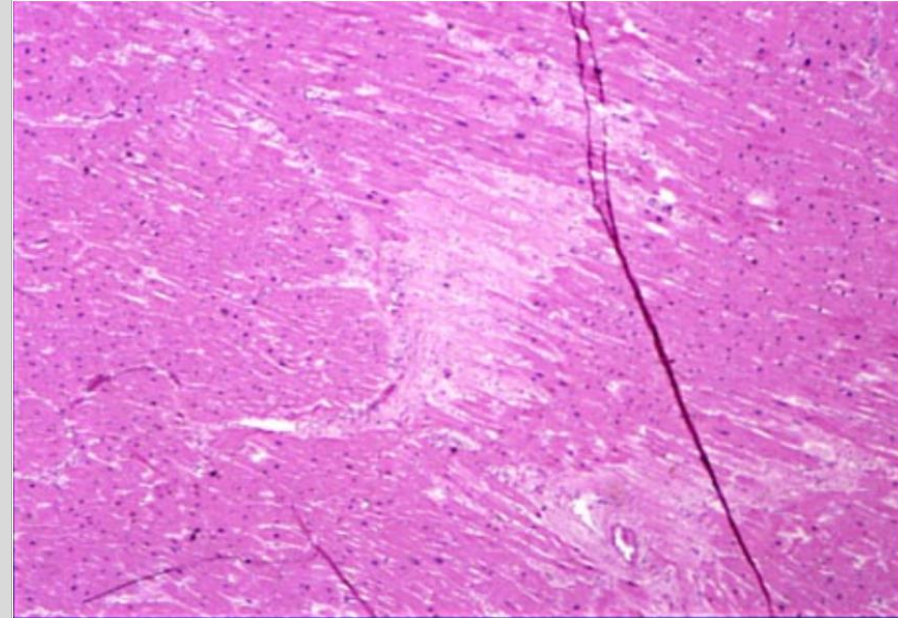
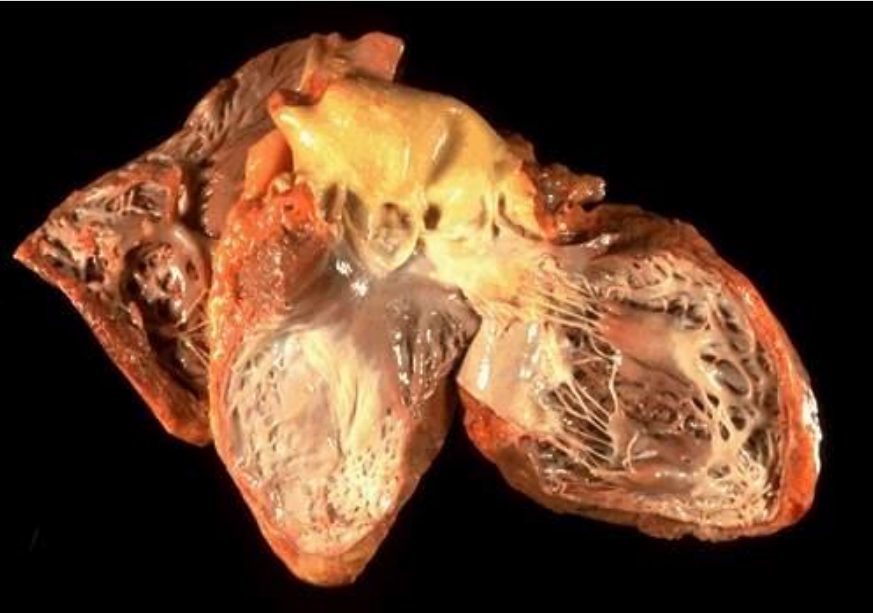


Acute MI: 2 weeks

- Healing MI
- Increased collagen, decreased cellularity or vascularity
- Grey-white scar, from border to center of infarct



Acute MI: 2 months



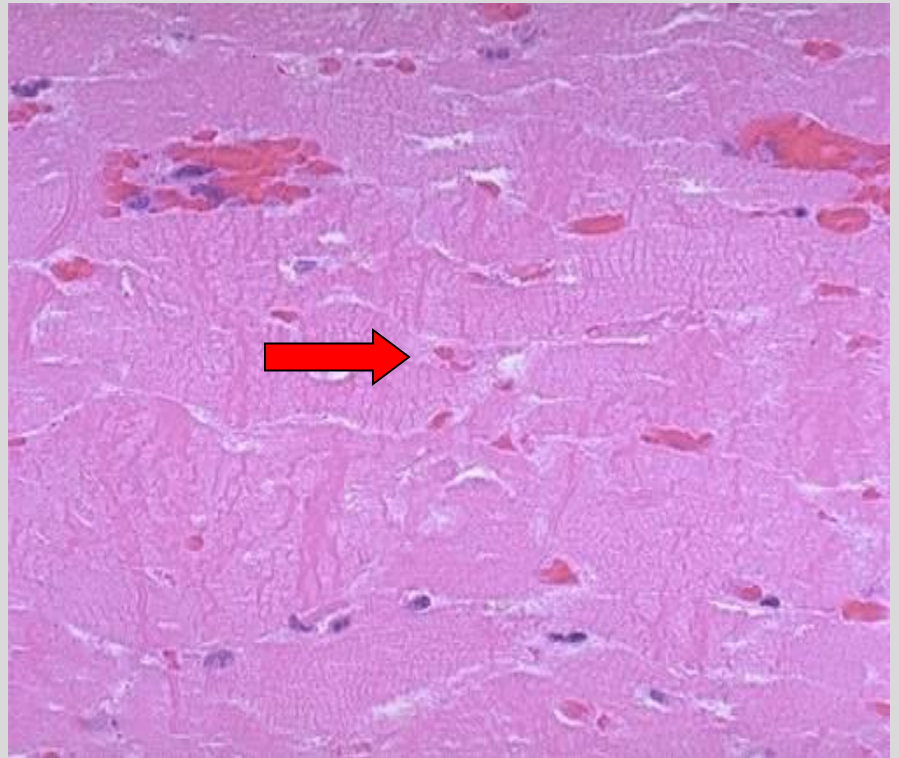
- Dense collagen scar

CONTRACTION BANDS

- Contraction bands - very eosinophilic transverse bands made of closely packed hypercontracted sarcomeres.
- Associated with **reperfusion injury**
- In myocardial infarction, most likely seen at the margin of infarct.
- Represent the effects of hypercontraction due to massive calcium influx across plasma membranes which increase actinomyosin interactions
- In the absence of ATP, sarcomeres cannot relax and “get stuck” in agonal tetanic state
- Contributing factors to reperfusion injury include mitochondrial dysfunction, myocyte hypercontracture, free radicals, leukocyte aggregation, platelet and complement activation

Contraction Bands

- The myocardial fibers are beginning to lose cross striations; nuclei not clearly visible in most cells.
- Many **irregular darker pink wavy contraction bands** extend across the fibers.
- Reperfusion



Complications of MI (gross and microscopic)

- Cardiogenic shock (massive infarction), Congestive heart failure (Low EF)
- **Arrhythmia** (conduction disturbances and myocardial irritability)
- **Acute fibrinous Pericarditis** presents as chest pain with friction rub
- **Rupture** of Ventricular free wall leads to cardiac tamponade
- **Rupture** of Septum (leads to LV->RV shunt)
- **Rupture of papillary muscle** – severe, acute mitral regurgitation
- **Ventricular Aneurysm - Mural thrombus: combo of local myocardial contractility abnormality (causing stasis) with endocardial damage (thrombogenic surface)**
- **Dressler syndrome** - autoimmune pericarditis

Question

- A 55 year old man has sudden onset of severe substernal chest pain that radiates to the neck. On PE, he is afebrile but has tachycardia, hyperventilation, and hypotension. Auscultation reveals no cardiac murmurs. The patient undergoes emergent coronary angiography which shows a thrombotic occlusion of the left circumflex artery and 50 to 70 % narrowing in the proximal circumflex and anterior descending arteries. Which of the following complications of this disease is most likely to occur within one hour of these events?
- A. ventricular fibrillation
- B. Pericarditis
- C. Myocardial rupture
- D. Ventricular aneurysm
- E. Thromboembolism

Arrhythmias

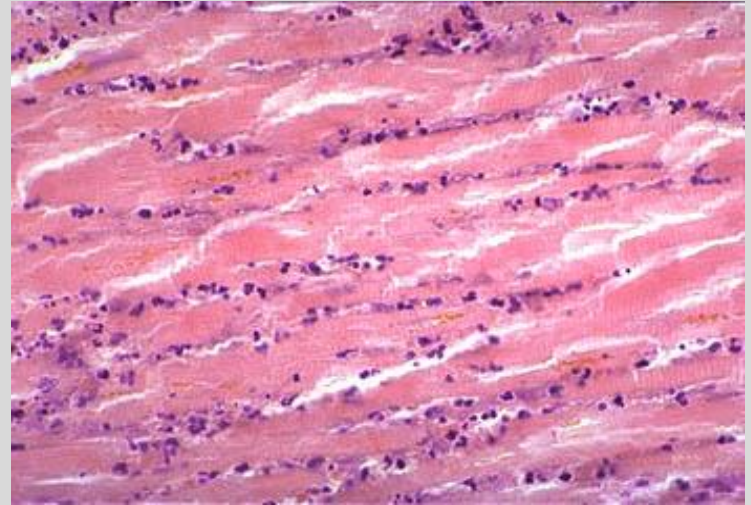
- Major source of pre-hospital mortality
- Anatomic interruption of perfusion to structures of the conduction pathway
- Toxic metabolic products & abnormal transcellular ion concentrations
- Scar tissue
- Autonomic stimulation
- Usually occurs 4-24 hours post MI

Arrhythmias

SA Node	RCA (70% of patients)
AV Node	RCA (85%)
Bundle of His	LAD (septal branches)
RBB	LAD (proximal portion) RCA (Distal portion)
LBB Left anterior fascicle Left posterior fascicle	LAD LAD and PDA

Acute Pericarditis

- Neutrophils are recruited to the site and degrade **dying cells and ECM**.
- They release **metalloproteinases** that degrade collagen and **irritate the pericardium** leading to acute fibrinous pericarditis
- Occurs within the **first 2 - 3 days**; **gradually resolves over next few days**



Pericarditis Clinical and Lab Findings

- Pericardial effusion- Muffled heart sounds
- Hypotension associated with pulsus paradoxus(abnormal drop in systolic blood pressure greater than 10mm Hg during inspiration)
- Neck vein distention
- CXR- “Water bottle” configuration
- Pericardial friction rub
 - Scratchy 3 component friction rub, best heard with the patient leaning forward

ECG findings show diffuse ST segment elevation except VR and V1 and PR segment depression

Commonly presents with sharp pain, aggravated by inspiration and relieved by sitting up and leaning forward



Dressler' s Syndrome

- Autoimmune induced fibrinous pericarditis occurring anywhere between **2nd week and 2nd month** post MI
- Autoantibodies directed against damaged pericardial antigens
- Similar symptoms as acute pericarditis

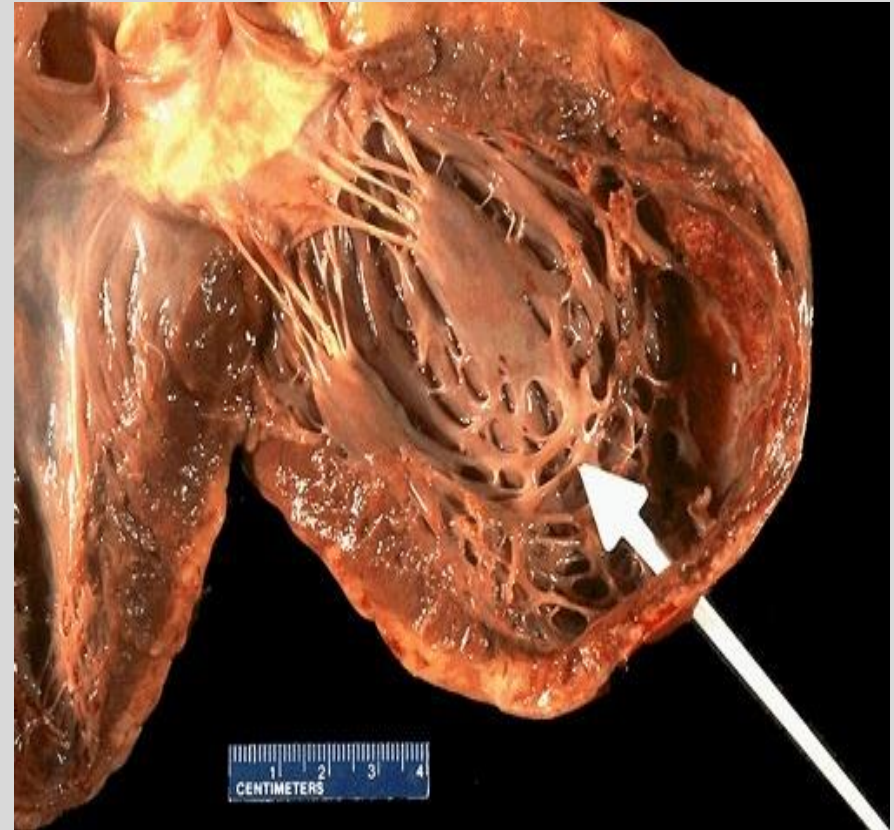
Myocardial Rupture

- Myocardial rupture, most likely between **3-7 days following MI**, when myocardium is softest – coagulative necrosis, neutrophilic infiltrate, and lysis of myocardial connective tissue weaken the infarcted myocardium
- Arrow marks point of rupture in this anterior-inferior MI of left ventricular free wall and septum



Rupture: Ventricular Free Wall

- Cardiac Tamponade - compression of heart by fluid (blood) in pericardium leading to decrease cardiac output
- Equilibration of diastolic pressure in all 4 chambers
- **Findings: Becks Triad** (hypotension, distend neck veins, distant heart sound)
- Increase in heart rate, **Pulsus Paradoxus**, Kussmaul sign (paradoxical rise in JVP on inspiration – indicates limited right ventricular filling)
- **EKG**: low voltage QRS and Electrical Alternans

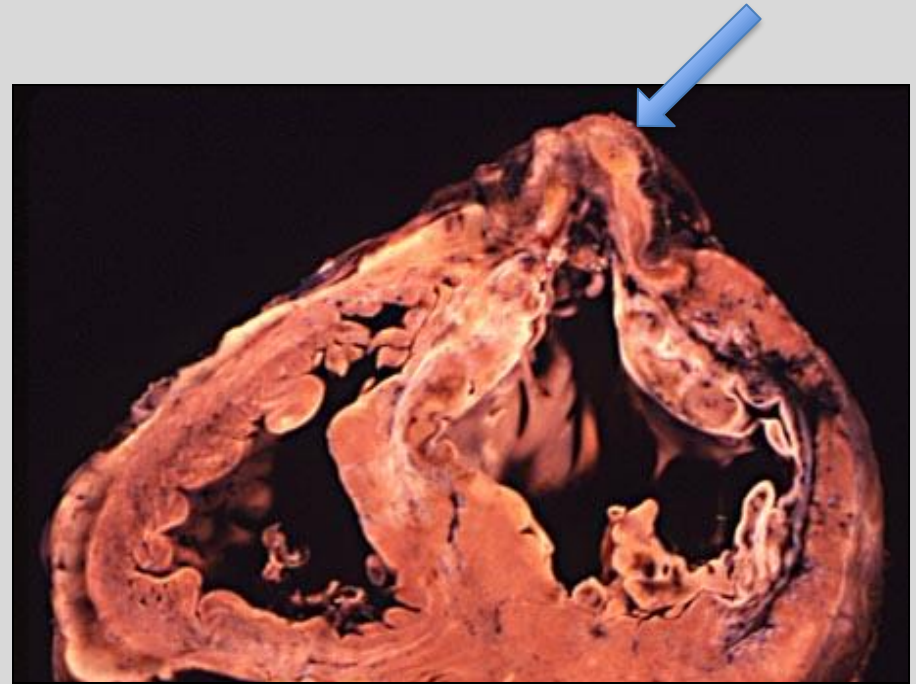


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False Aneurysm

- What if the rupture is incomplete?
- LV free wall held in check by thrombus that plugs the hole in the myocardium
- Epicardium and adherent parietal pericardium form “wall”



Question

- A 58 year old man suffered a myocardial infarction involving the apex of the left ventricle. Six months later, an echocardiogram shows the development of a ventricular bulge that does not contract during systole. The patient subsequently suffers a massive stroke and suddenly dies. Which of the following would be an expected finding at autopsy?
- A. Calcific aortic stenosis
- B. Dilated cardiomyopathy
- C. Mitral valve prolapse
- D. mural thrombus
- E. Ventricular rupture

True Ventricular Aneurysm

- Cross section through the heart reveals a ventricular aneurysm with a very thin wall at the arrow
- Note how the aneurysm bulges out
- The stasis in this aneurysm allows mural thrombus to form within the aneurysm.

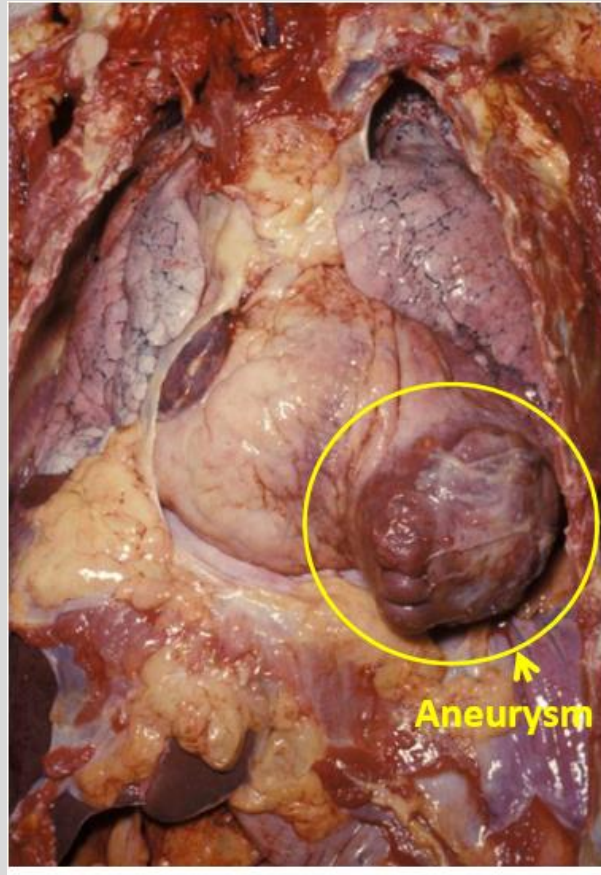


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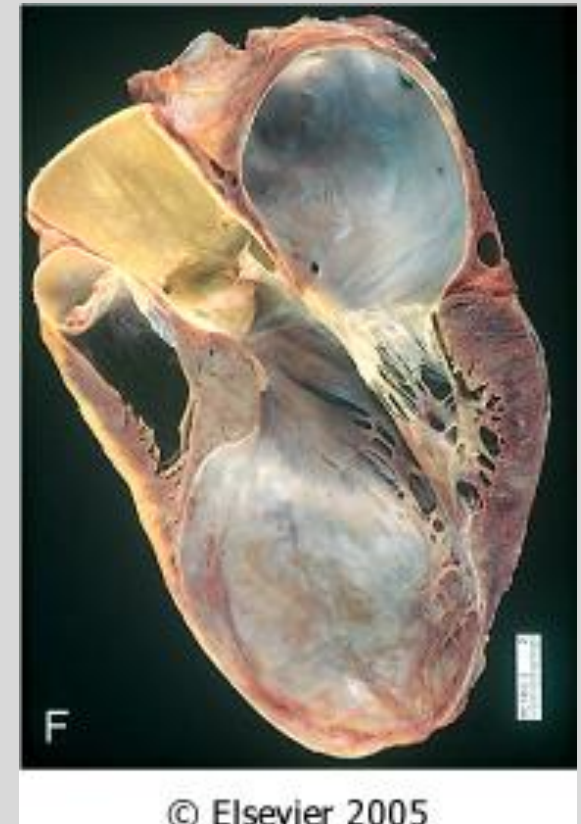
True Ventricular Aneurysm

- Late complication of MI: weeks – months after
- Does NOT involve communication between the LV and the pericardium
- Usually does not rupture



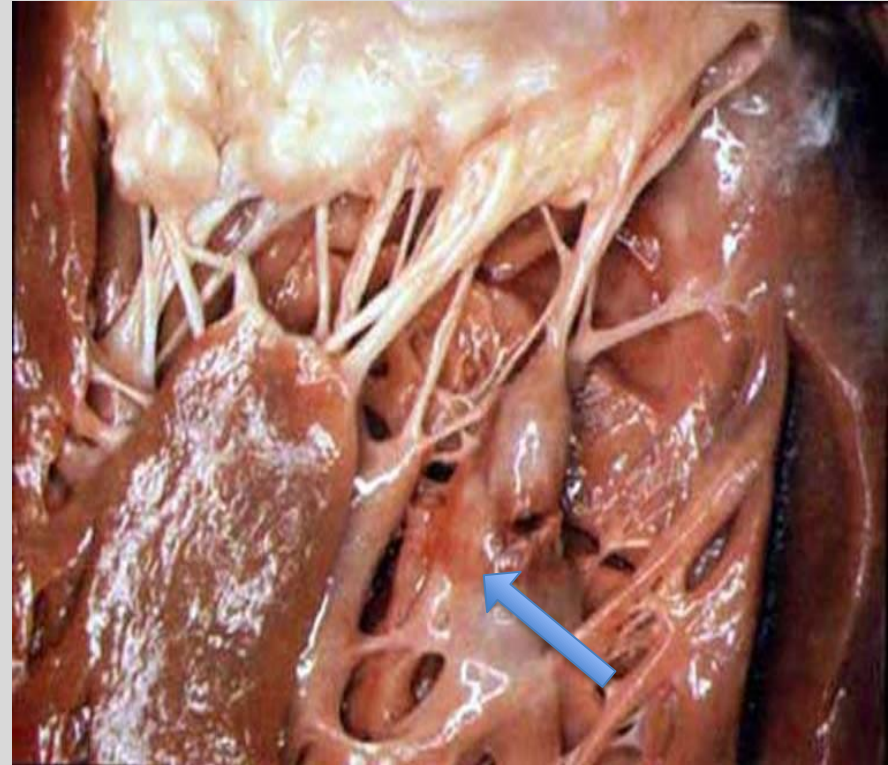
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Rupture: Papillary Muscle Rupture

- Leads to incompetent mitral valve (MR)
- Clinical Presentation: Blood backs up into lungs, transudation of fluid, shortness of breath, pansystolic murmur.
- Posteromedial LV papillary muscle most susceptible b/c of its more precarious blood supply
- Occlusion of RCA, in right dominant heart pt., ischemia to posterior LV 1/3 septum - where posterior leaflet of mitral valve attaches.

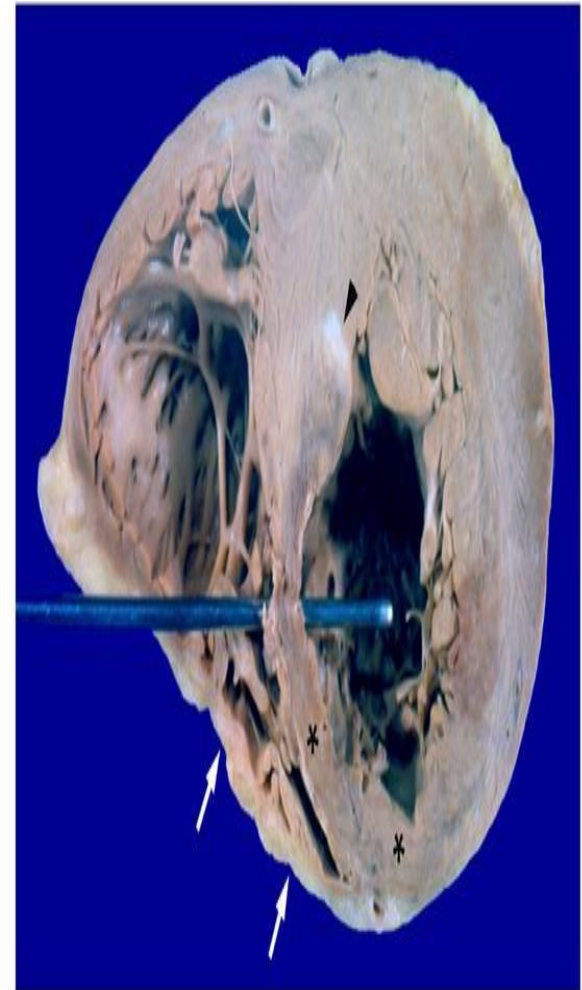


Rupture: Ventricular Septum

- Analogous to LV free wall rupture, but consequence and patient presentation are different
- Blood shunted LV → RV
- Pt presents with Right sided heart failure, JVD, pedal edema
- Loud systolic murmur at left sternal border, representing transeptal flow
- Most often associated with the LAD coronary artery thrombosis

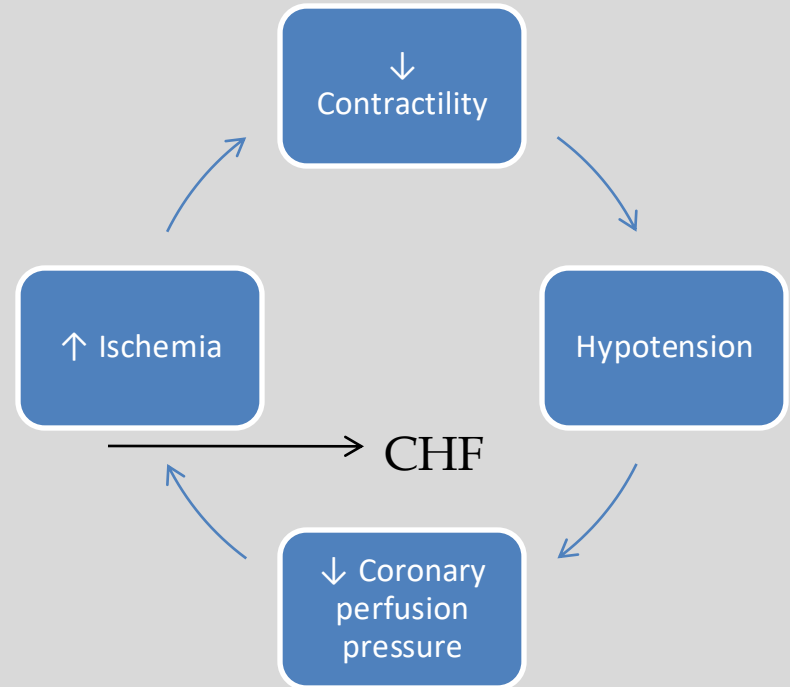
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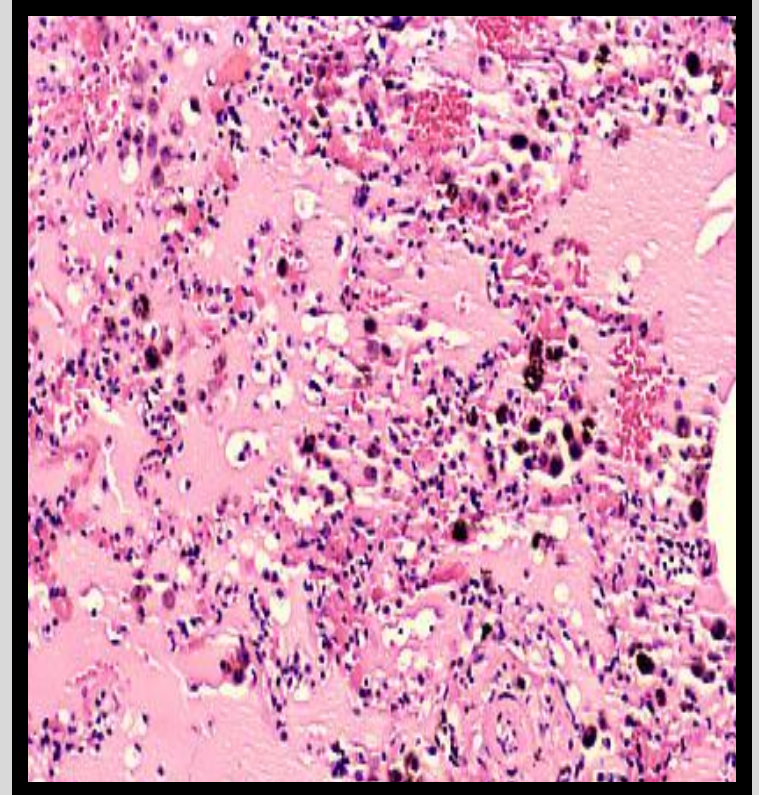
Contractile Dysfunction

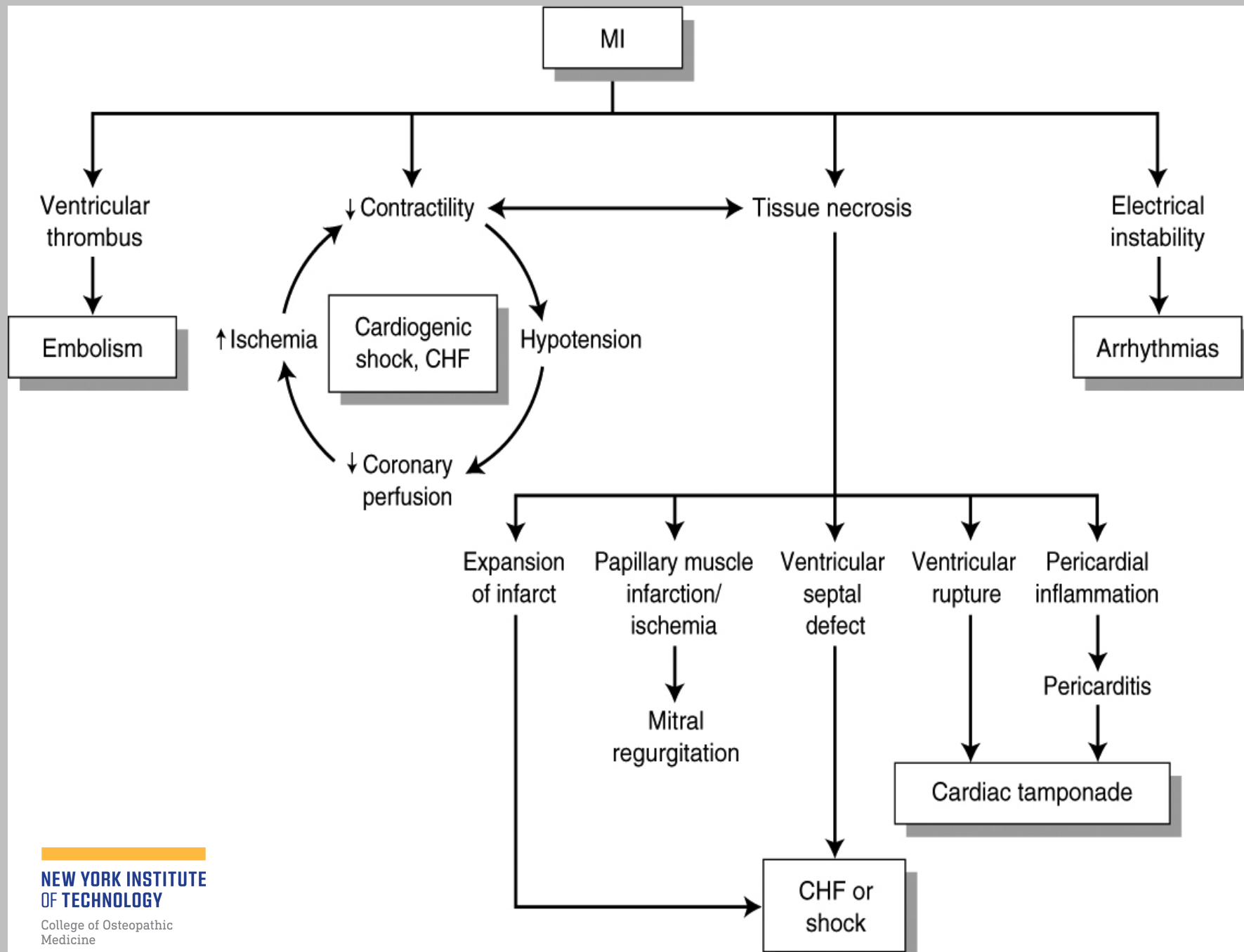
- Congestive heart failure
 - Left ventricular failure
- Cardiogenic shock - severely decreased cardiac output and hypotension
- Inadequate perfusion of peripheral tissues: develops when more than 40% of the LV has infarcted



Pulmonary Edema and Congestion with Heart Failure Cells

- Blood is phagocytosed by alveolar macrophages
- **Hemosiderin accumulates in alveolar macrophages called heart failure cells**





MI Treatment

- Aspirin and or heparin – limits thrombosis
- Supplemental oxygen – minimize ischemia
- Nitrates – vasodilate coronary arteries (decrease preload = decrease stress)
- Beta-Blockers – slows heart rate, decreasing oxygen demand and risk for arrhythmia
- ACE Inhibitors – decrease LV dilation
- Fibrinolysis or angioplasty – opens blocked vessel

Question

- A 52 year old man with a history of diabetes mellitus and hypertension experiences **pain in his left arm and shoulder** for the past 2 hours. Over the next 6 hours he develops shortness of breath which continues for 2 days. On **day 3** he sees his doctor who notes a normal temperature, a pulse of 82 bpm, respirations of 18 bpm, and blood pressure of 160/100 mm Hg. Lab studies at this time show that **creatinine kinase serum levels are within normal** limits but **troponin I level is increased**. The patient is admitted to the hospital and continues to have dyspnea for the next 3 days. On **day 7 after the initial onset of shoulder** pain, the patient has a **cardiac arrest** and does not survive resuscitative efforts. An autopsy performed shows a large **transmural infarction** of the left anterior free wall with **rupture** and hemopericardium. Which statement is most likely true?

Answer Choices

- A. Myocardial infarction did not develop until day 5 or 6 after the episode of shoulder pain
- B. The normal CK level on day 3 excludes the possibility of myocardial infarction within the preceding 72 hours
- C. the patient had an acute myocardial infarction on the day he developed shoulder pain
- D. A CK-MB level would have detected an acute MI on day 3
- E. The patient had a second MI which caused a myocardial rupture on day 7

Review Questions

- What type of necrosis that occurs following MI?
- What type of cells infiltrate the myocardium 1-3 days post-MI and 3-7 days post MI ?

Summary Slide

- Myocardial infarction – definition
- Subendocardial v. Transmural
- Frequency and sites
- EKG changes
- Time sequence changes – see text, power point slide for details
 - 0-30 minutes reversible no real change
 - 30 min – 4 hrs “wavy fibers”
 - 4 – 12 hrs gross: focal mottling **HISTO: edema, start coag necrosis**
 - 12 – 24 hrs: gross: dark mottling **HISTO: coag necrosis, pyknosis, contraction bands, start PMNs**
 - 1 – 3 days: gross: yellow, tan center **HISTO: neutrophil infiltration**
 - 3 – 7 days: gross: hyperemic border, central yellow-tan **HISTO prominent macrophages, beginning granulation tissue**
 - 7 – 10 days: **HISTO: granulation tissue prominent**
 - 2 weeks: **HISTO: increased collagen formation**
 - 2 months – **HISTO: well formed scar**
- Laboratory tests/clinical correlation
- Complications – arrhythmias, pericarditis (acute fibrinous), pericarditis (Dressler syndrome), ventricular free wall rupture, false v. true aneurysm, papillary muscle rupture, ventricular septal rupture, contractile dysfunction

Reference

- Robbins and Cotran Pathologic Basis of Disease, 10th Edition, 2020
- Douglas L. Mann, Murali Chakinala. Harrison's Principles of Internal Medicine. 19th edition.
- First Aid for USMLE

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



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